

HCAI5TML01 – Mathematics of Learning.
Theme: Getting Started with Learning Theory.
Lec02 – From Averages to **Guarantees:**
Inequalities in Probability

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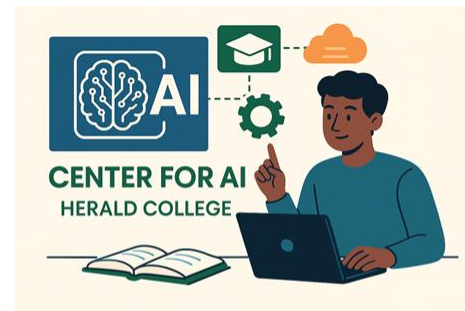


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Where we left

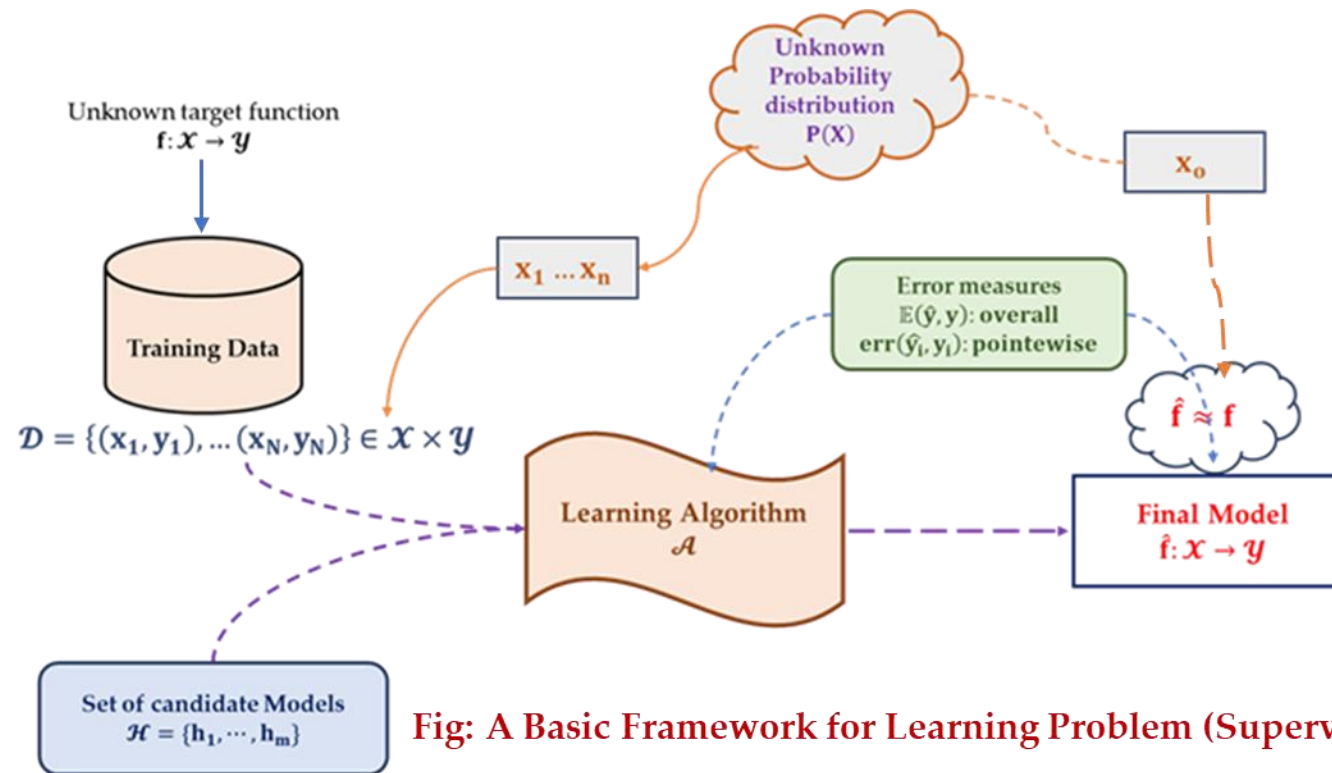
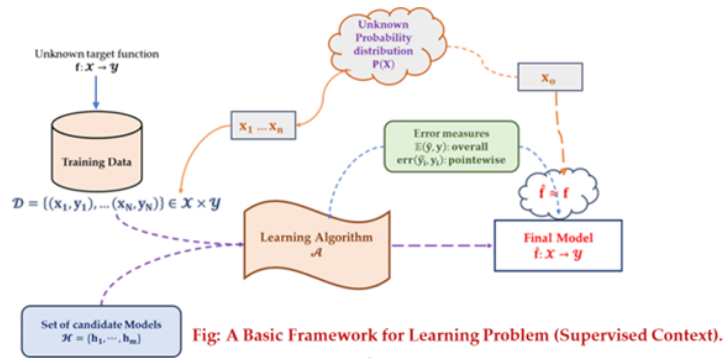


Fig: A Basic Framework for Learning Problem (Supervised Context).

Where we left



ERM – Framework.

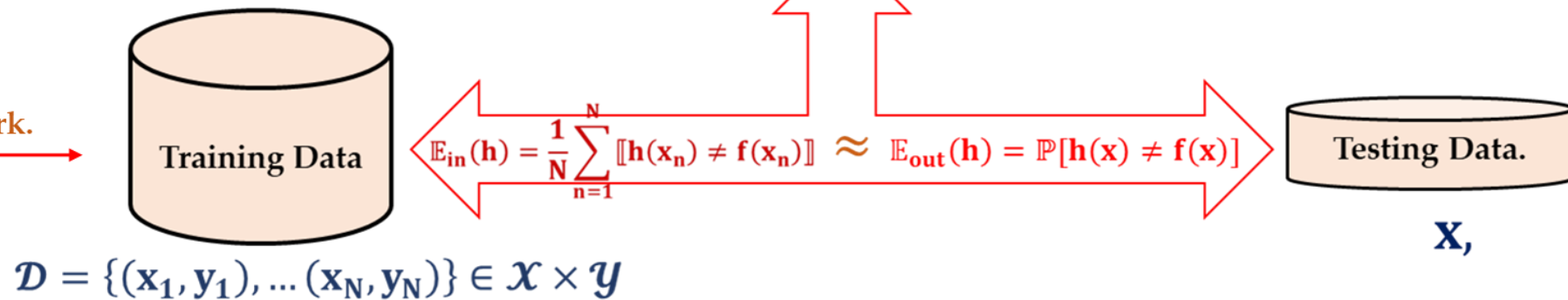


Fig: $\mathbb{E}_{\text{in}}$ is evaluated using the training data, whereas $\mathbb{E}_{\text{out}}$ is evaluated using the testing sample.

Where are we heading ...

- In ML, we only observe **training error** (empirical).
 - But we care about **test error** (true/generalization).
- In **Learning Theory** Terms:
 - The **empirical risk (training error)** is an average over the finite sample.
 - The **true risk (generalization error)** is over the whole population distribution.
- Big Q:
 - How do we go from *what we see in the training data (averages)* to *confidence about the true error or behavior (guarantees)* ?"

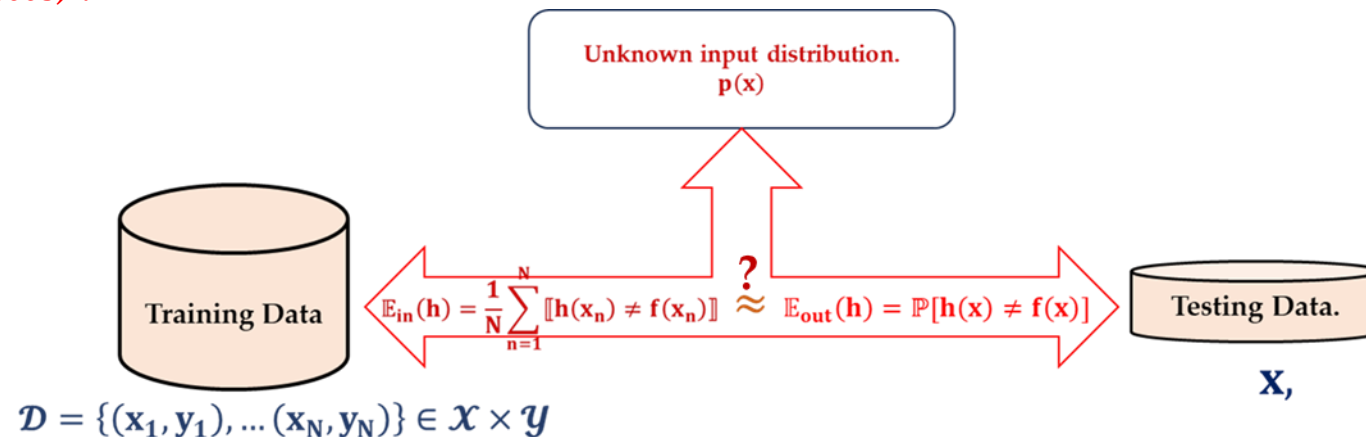


Fig: $\mathbb{E}_{in}$ is evaluated using the training data, whereas $\mathbb{E}_{out}$ is evaluated using the testing sample.

Observed Data (**Empirical Averages**) → Confidence Statements about Population (**Theoretical Guarantees**).

In – Equalities in Probability.

{1. Markov's Inequality.}

1.1 Markov's Inequality – Intuition.

- It is an elementary inequality that **links probability** and **expectation**.
- It says:
 - If the **average (Expectation – expected)** value of something is small,
 - then **the chance (probability)** of it being **much larger than** that **average** is also small.
- For example:
 - If the **expected number of daily complaints is 2**,
 - then **the probability of getting 10 or more complaints is at most $\frac{2}{10} = 0.2$** .
 - Even without knowing the distribution of complaints, this still holds.

1.2 Markov's Inequality – Statement.

Theorem 1 Markov's Inequality:

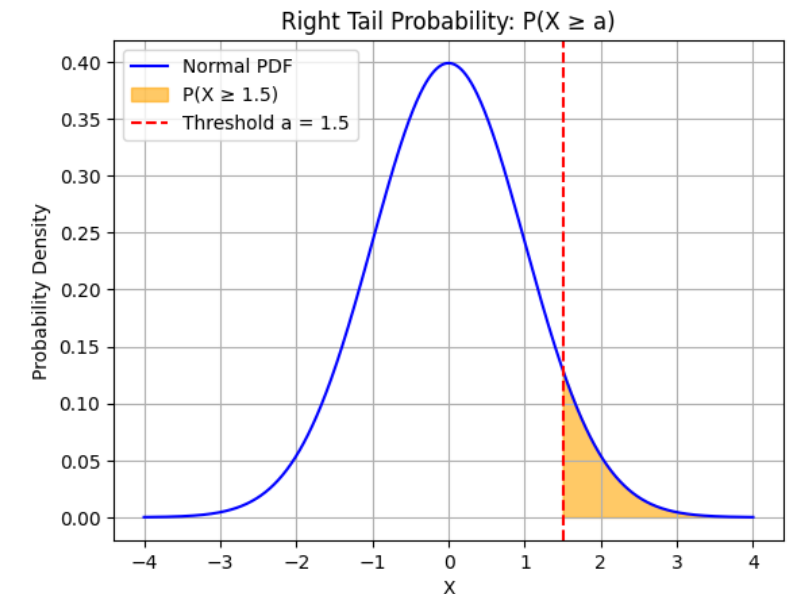
Let $X \geq 0$ be a non-negative random variable. Then, for any $\epsilon > 0$, we have

$$P[X \geq \epsilon] \leq \frac{E[X]}{\epsilon}.$$

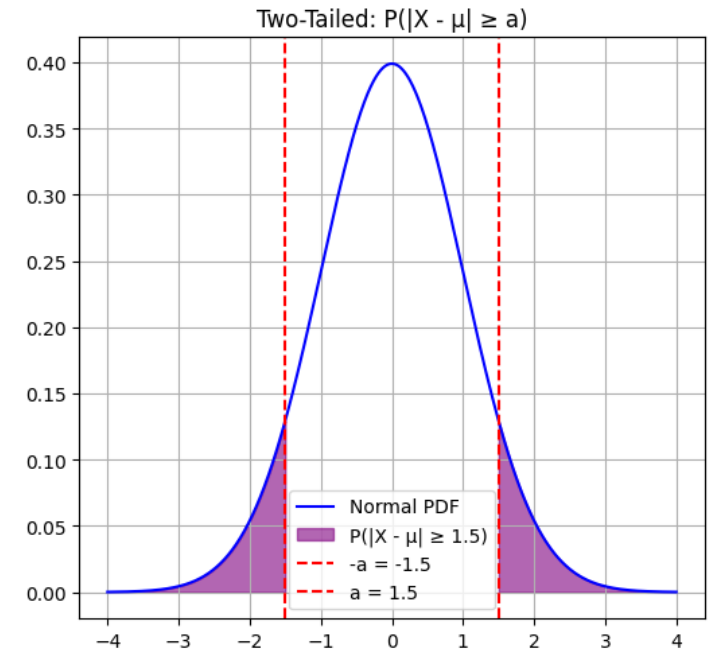
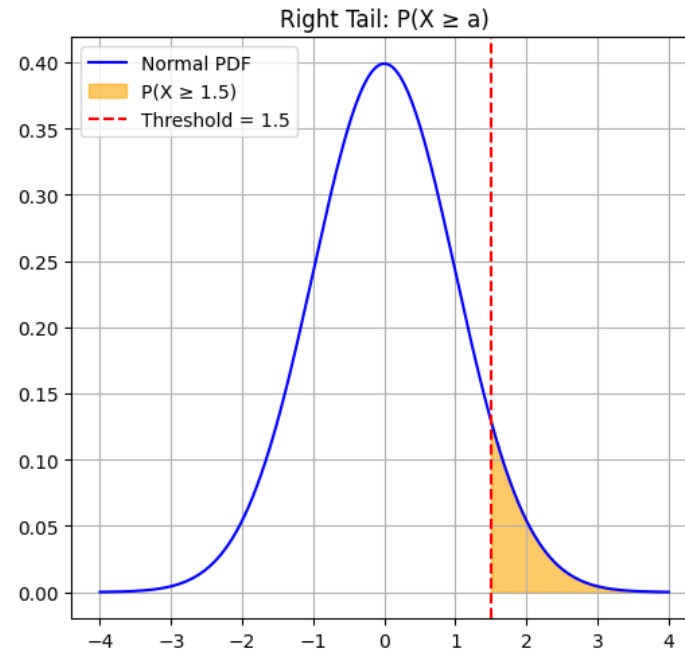
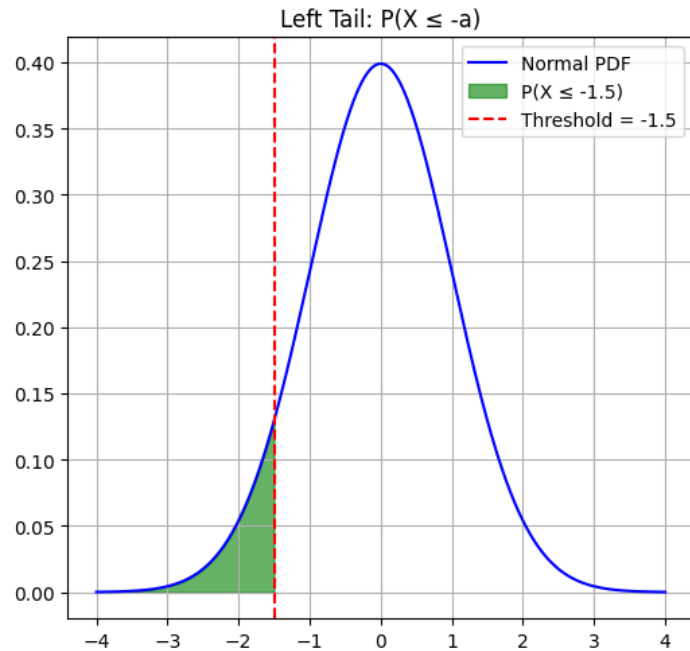
- Markov's Inequality helps bound the **tail probability** of
 - a **non – negative random variable using only its expectation.**
- Suppose we **don't know** the **full distribution of X** , or it is **too complex to analyze**.
- But you do know **the expected value $E[X]$** .
- Markov inequality tells you:
 - the probability **that X exceeds a threshold ϵ is at most $\frac{E[X]}{\epsilon}$.**
- Markov gives a **loose but general** bound
 - without needing any info about the variance or shape of the distribution.

1.3 Proof – Markov's Inequality.

- Before – Tail Probability:
 - Tail probability refers to
 - the probability that a random variable takes on values far from its mean,
 - specifically values in the “tails” of its distribution.
 - Formally:
 - For a random variable X and a threshold a , the **right tail** probability is:
 - $P(X \geq a)$
 - This is the probability that x takes a value at least as large as a .
 - Similarly, the **left tail** probability is:
 - $P(X \leq a)$
 - In the context of Markov or Chebyshev, tail probability usually entail **right tail**.



1.3.1 Tail Probability – Demonstration.



1.3.2 Formal Proof.

Setup:

- X is a random variable such that $X \geq 0$ (non-negative).
- $a > 0$ is a fixed positive number (threshold).
- We want to bound the probability that X is at least a , i.e., $P(X \geq a)$.

1.3.2 Formal Proof.

Step 1: Split the expectation using indicator functions:

Intuition: We break the average value of $\mathbf{X}$ into two parts:

- One where $\mathbf{X}$ is smaller than $\mathbf{a}$
- One where $\mathbf{X}$ is at least $\mathbf{a}$

Write the expectation as the sum over two disjoint events:

$$\mathbf{E}[\mathbf{X}] = \mathbf{E}[\mathbf{X} \cdot \mathbf{1}_{\mathbf{X} < \mathbf{a}}] + \mathbf{E}[\mathbf{X} \cdot \mathbf{1}_{\mathbf{X} \geq \mathbf{a}}]$$

where:

- $\mathbf{1}_{\mathbf{X} < \mathbf{a}}$ is the indicator function that is 1 if $\mathbf{X} < \mathbf{a}$, else 0
- $\mathbf{1}_{\mathbf{X} \geq \mathbf{a}}$ is the indicator function for the event $\mathbf{X} \geq \mathbf{a}$.

1.3.2 Formal Proof.

Step 2: Drop the first term (non-negative) to get a lower bound:

Because $\mathbf{X} \geq 0$, both terms are non-negative, so:

$$\mathbf{E}[\mathbf{X}] \geq \mathbf{E}[\mathbf{X} \cdot \mathbf{1}_{\mathbf{X} \geq \mathbf{a}}]$$

Intuition: The expected value considering only the part where $\mathbf{X} \geq \mathbf{a}$ is less than or equal to the total expected value.

Step 3: Lower bound on the integrand:

Use the fact that on the event $\mathbf{X} \geq \mathbf{a}$, the value of $\mathbf{X}$ is at - least $\mathbf{a}$.

By definition:

$$\mathbf{X} \cdot \mathbf{1}_{\mathbf{X} \geq \mathbf{a}} \geq \mathbf{a} \cdot \mathbf{1}_{\mathbf{X} \geq \mathbf{a}}$$

Because:

- when $\mathbf{X} \geq \mathbf{a}$, multiplying by the indicator keeps $\mathbf{X}$ and $\mathbf{X}$ is at least $\mathbf{a}$.
- When $\mathbf{X} < \mathbf{a}$, the indicator is 0, so inequality trivially holds.

1.3.2 Formal Proof.

Step 4: Take expectations of the lower bound

Replace $\mathbf{X} \cdot \mathbf{1}_{\mathbf{X} \geq \mathbf{a}}$ with $\geq \mathbf{a} \cdot \mathbf{1}_{\mathbf{X} \geq \mathbf{a}}$ inside expectation:

$$E[X \cdot \mathbf{1}_{X \geq a}] \geq E[a \cdot \mathbf{1}_{X \geq a}] = a \cdot E[\mathbf{1}_{X \geq a}]$$

Step 5: Expectation of indicator is probability:

Recognize $\mathbf{E}[\mathbf{1}_{\mathbf{X} \geq \mathbf{a}}]$ is the probability of the event $\mathbf{X} \geq \mathbf{a}$

$$\mathbf{E}[\mathbf{1}_{\mathbf{X} \geq \mathbf{a}}] = \mathbf{P}(\mathbf{X} \geq \mathbf{a})$$

Since the indicator is 1 when $\mathbf{X} \geq \mathbf{a}$ and 0 otherwise, the expectation of the indicator equals the probability of the event.

1.3.2 Formal Proof.

Step 6: Combine all inequalities:

Putting together steps 2, 4, and 5:

$$\mathbf{E}[\mathbf{X}] \geq \mathbf{a} \cdot \mathbf{P}(\mathbf{X} \geq \mathbf{a})$$

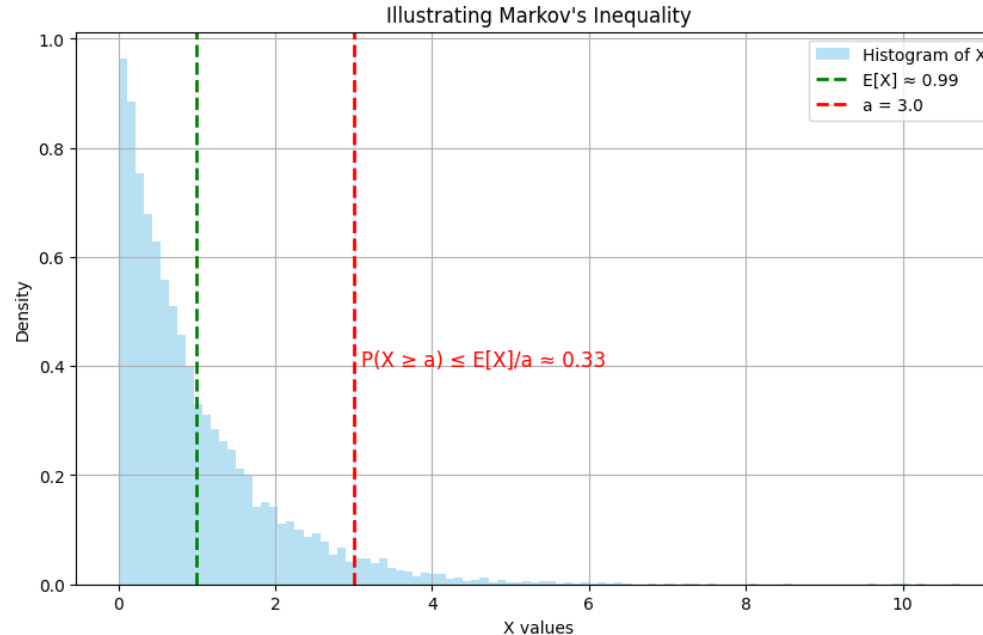
Final conclusion: Markov's inequality

Divide both sides by **a** (positive, so inequality direction is preserved):

$$\mathbf{P}(\mathbf{X} \geq \mathbf{a}) \leq \frac{\mathbf{E}[\mathbf{X}]}{\mathbf{a}}$$

This shows that the probability of **X** being large ($\geq \mathbf{a}$) is at most the ratio of the mean of **X** to **a**.
This is the **Markov Inequality**.

1.4 What does it mean?



Markov Bound:

- Markov's inequality provides an **upper bound** on the probability that
 - a **non-negative random variable** is **greater than or equal to some positive constant**.
- In simpler terms, it **gives us a maximum possible value** for the probability $P(X \geq a)$, based on the **expected value (mean)** of X .
- In the plot, the **calculated Markov bound is shown in red text**.
 - Notice that the empirical probability (the actual probability calculated from the simulated data)
 - is less than the Markov bound, which is consistent with the inequality.

1.5 Example.

Example:

Let $\mathbf{X} \sim \text{Uniform}(\mathbf{0}, \mathbf{4})$. We verify Markov's inequality for different thresholds:

Solution. First, we observe that:

$$\mathbf{E}[\mathbf{X}] = \frac{\mathbf{0} + \mathbf{4}}{\mathbf{2}} = \mathbf{2}$$

Threshold	Actual Probability	Markov's Bound	Comparison
$\mathbf{P}[\mathbf{X} \geq \mathbf{2}]$	$\frac{\mathbf{4} - \mathbf{2}}{\mathbf{4}} = \mathbf{0.5}$	$\frac{\mathbf{E}[\mathbf{X}]}{\mathbf{2}} = \mathbf{1}$	$\mathbf{0.5} \leq \mathbf{1}$
$\mathbf{P}[\mathbf{X} \geq \mathbf{3}]$	$\frac{\mathbf{4} - \mathbf{3}}{\mathbf{4}} = \mathbf{0.25}$	$\frac{\mathbf{E}[\mathbf{X}]}{\mathbf{3}} \approx \mathbf{0.\overline{6}}$	$\mathbf{0.25} \leq \mathbf{0.\overline{6}}$
$\mathbf{P}[\mathbf{X} \geq \mathbf{4}]$	$\frac{\mathbf{4} - \mathbf{4}}{\mathbf{4}} = \mathbf{0}$	$\frac{\mathbf{E}[\mathbf{X}]}{\mathbf{4}} = \mathbf{0.5}$	$\mathbf{0} \leq \mathbf{0.5}$

Conclusion:

- While Markov's inequality provides valid upper bounds in all cases ($\mathbf{P}[\mathbf{X} \geq \mathbf{a}] \leq \frac{\mathbf{E}[\mathbf{X}]}{\mathbf{a}}$),
- the bounds become increasingly loose as the threshold $\mathbf{a}$ grows larger.
- For this uniform distribution, the actual probabilities are significantly smaller than Markov's bounds.

In – Equalities in Probability.

{2. Chebyshev's Inequality.}

2.1 Intuition Behind Chebyshev's Inequality.

- **Chebyshev's inequality** gives a **universal guarantee** about **how spread** out a **random variable** can be,
 - no matter **what its distribution is**, as long as it has a **finite mean and variance**.
- Imagine you are blindfolded and only told:
 - The average value (mean) of a variable.
 - How spread out it is (standard deviation).
 - You don't know the shape of the distribution:
 - it could be skewed, lumpy, or anything else.
 - **Still Chebyshev says:**
 - No matter **what the distribution** looks like,
 - at least $1 - \frac{1}{k^2}$ of the values will lie **within k standard deviations of the mean**.

2.1.1 Examples to Build Intuition.

k	Max Prob. Outside $\mu \pm k\sigma$	Max Prob. Inside $\mu \pm k\sigma$
1	≤ 1.00	$\geq 0\%$
2	≤ 0.25	$\geq 75\%$
3	$\leq 0.111 \dots$	$\geq 88.9\%$
4	≤ 0.0625	$\geq 93.75\%$

- So even without knowing anything else:
 - At least 75% of the values must lie within $\mu \pm 2\sigma$
 - At least 88.9% of the values must lie within $\mu \pm 3\sigma$
- That's powerful, because it's distribution free.
- Example:
 - For **detecting outliers**: if something is more than 3σ away, it's likely rare.

2.2 Chebyshev's Inequality – Statement.

- Let **X be a random variable** with **finite mean** $\mu = \mathbb{E}[X]$ and **finite non – zero variance** $\sigma^2 = \text{Var}(X)$.
 - Then for any **k > 0**:
 - $\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$
 - or equivalently,
 - $\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$
- **Interpretation:**
 - This inequality tells us that **no more than** $\frac{1}{k^2}$ of the **probability mass** lies **more than k standard deviations from the mean**.
 - It applies **to any distribution** with **finite mean and variance** (not necessarily normal).
 - For example,
 - $\mathbb{P}(|X - \mu| \geq 2\sigma) \leq \frac{1}{4}$,
 - meaning at least 75% of values are within 2 standard deviations of the mean.

2.3 Formal Proof.

Step - by - step Proof (Using Markov's Inequality)

We are interested in the probability that a random variable deviates from its mean by at least k standard deviations:

$$\mathbf{P}(|\mathbf{X} - \mu| \geq k\sigma)$$

. We apply Markov's Inequality to the non-negative random variable $\mathbf{Y} = (\mathbf{X} - \mu)^2$:

- Let $\mathbf{Y} = (\mathbf{X} - \mu)^2 \geq 0$
- Then: Squaring both sides of the inequality preserves the inequality (since both sides are ≥ 0), so:

$$|\mathbf{X} - \mu| \geq k\sigma \iff (\mathbf{X} - \mu)^2 \geq k^2\sigma^2$$

Thus:

$$\mathbf{P}(|\mathbf{X} - \mu| \geq k\sigma) = \mathbf{P}((\mathbf{X} - \mu)^2 \geq k^2\sigma^2)$$

- Apply **Markov's Inequality** to $\mathbf{Y}$:

$$\mathbf{P}((\mathbf{X} - \mu)^2 \geq k^2\sigma^2) \leq \frac{\mathbf{E}[(\mathbf{X} - \mu)^2]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

- Hence proved:

$$\mathbf{P}(|\mathbf{X} - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

✓ Chebyshev's Inequality

2.4 Example

Example:

Let $\mathbf{X} \sim \text{Uniform}(\mathbf{0}, 2\sqrt{2})$. Find the bound of Chebyshev's inequality for the probability

$$\mathbf{P}[|\mathbf{X} - \mu| \geq \mathbf{1}].$$

2.4 Example

Note that:

- $E[X] = \frac{0+2\sqrt{2}}{2} = \sqrt{2}$
- $\sigma^2 = \frac{(2\sqrt{2}-0)^2}{12} = \frac{8}{12} = \frac{2}{3}$

Applying Chebyshev's inequality with $\varepsilon = 1$:

$$P[|X - \mu| \geq 1] \leq \frac{\sigma^2}{\varepsilon^2} = \frac{\frac{2}{3}}{1} = \frac{2}{3}$$

Note: While $\frac{2}{3}$ is a valid upper bound, it's not particularly informative since we already know the exact probability must be less than 1 for any non-trivial case. The uniform distribution actually gives:

$$P[|X - \sqrt{2}| \geq 1] = 1 - \frac{2(\sqrt{2} - 1)}{2\sqrt{2}} \approx 0.2929$$

showing Chebyshev's bound is **quite conservative** in this case.

2.5 Why is Chebyshev's Inequality Conservative?

- When we say that **Chebyshev's inequality** provides a "conservative" bound,
 - it means that while **the inequality is mathematically correct** and **always valid**,
 - it often gives an **upper bound that is much larger** than the actual probability.
- Why?
 1. **Works for Any Distribution:**
 - **Chebyshev's inequality** applies to **any distribution** (with finite mean and variance),
 - regardless of its shape.
 - Thus, it must **cover all possible cases**,
 - including the **worst-case scenarios**,
 - the bound tends to **be loose for well behaved distributions** (e.g. uniform, normal).
 2. **Does Not Use Distribution – specific Information:**
 - Unlike **exact probability calculations** (which depend on the PDF/CDF of X),
 - **Chebyshev's inequality** only **uses the variance σ^2** .
 - This means it **ignores** the **actual shape** of the distribution,
 - leading to a more **"pessimistic"** estimate.

2.5.1 For our example:

Chebyshev's Inequality Bound

For $\mathbf{X} \sim \text{Uniform}(0, 2\sqrt{2})$:

$$P(|\mathbf{X} - \sqrt{2}| \geq 1) \leq \frac{\sigma^2}{\epsilon^2} = \frac{2/3}{1} = \frac{2}{3} \approx 0.6667$$

Exact Probability Calculation

Using the uniform distribution properties:

$$\begin{aligned} P(|X - \sqrt{2}| \geq 1) &= 1 - P(\sqrt{2} - 1 < X < \sqrt{2} + 1) \\ &= 1 - \frac{(\sqrt{2} + 1) - (\sqrt{2} - 1)}{2\sqrt{2}} \\ &= 1 - \frac{2}{2\sqrt{2}} \\ &\approx 1 - 0.7071 \\ &\approx 0.2929 \end{aligned}$$

Comparison Summary

Method	Bound/Probability
Chebyshev's Inequality	≤ 0.6667
Exact Probability	≈ 0.2929

Conclusion: Chebyshev's bound (≈ 0.6667) is more than twice as large as the true probability (≈ 0.2929), demonstrating that while mathematically correct, it can be quite conservative for well-behaved distributions like the uniform distribution.

Thank You